



Ethereum transaction tracking: Inferring evolution of transaction networks via link prediction

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ABSTRACT

Blockchain is an emerging technology which has attracted wide attention in recent years. As one of the blockchain applications, cryptocurrency has developed rapidly in recent years, attracting criminals to commit fraud and money laundering. Therefore, to better protect the legitimate interests of users and help formulate an effective supervision, it is necessary to track and follow transaction records on blockchain-based systems. This paper studies the problem of transaction tracking in Ethereum from a network perspective, aiming to study explainable strategies for money flow generation. We first collect the space-intensive transaction data from Ethereum blockchain and model them as temporal weighted multi-digraphs. A variety of tracking strategies considering different transaction factors (i.e., frequency and amount) are proposed, and the corresponding random-walk based link predictions method are designed for evaluation. Our method gets explainable results from the experiments, demonstrating that both transaction frequency and amount influence the generation of new transactions in Ethereum. This means when tracking the money flow among Ethereum accounts, we should pay more attention to those transaction paths having a shorter time interval and a larger amount. From these transaction features, the proposed random-walk based link prediction framework is found to be an effective method for transaction tracking. Furthermore, we show an application of transaction tracking via link prediction effectively enhance the ability to detect the suspicious accounts in Ethereum.

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1. Introduction

Blockchain is an open and distributed ledger technology managed by a peer-to-peer network via a consensus mechanism [1], and most blockchain systems inherit the characteristics of being decentralized, open, autonomous, as well as user tamperability and anonymity. As the representative of “blockchain 1.0”, Bitcoin is the first decentralized peer-to-peer cryptocurrency whose price hit an all-time high of 19,850 USD on December 17, 2017. Launched in 2015, Ethereum is currently the largest open-source blockchain-based platform that supports distributed applications (i.e., smart contracts) [2], and has been widely recognized as the core of “blockchain 2.0”. The corresponding cryptocurrency, *Ether*, is the digital asset of the account in Ethereum, and it is also the medium used to pay transaction fees. With its rapid development in the past few years, Ether has become the second largest cryptocurrency and its market capitalization

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reached around 20 billion USD in 2018. Attention has been paid to the modeling and understanding of the blockchain systems and corresponding cryptocurrencies of Bitcoin, Ethereum [3–12].

However, as blockchain projects continue to develop and more people invest, blockchain-related crimes have mushroomed. Taking advantage of the anonymity of blockchain, criminals attempt to evade supervision and engage in money laundering, smuggling and other illegal activities. For instance, Bitcoin is reported to be involved in several illegal transactions on the dark web marketplaces, e.g., “Silk Road”, which hid drug dealings, assassinations and other crimes [13]. Besides, in Ethereum, which supports the functionality of smart contracts, more types of cybercrimes have been reported, such as exploits, hacks, phishing, and Ponzi schemes [14].

The past few years have witnessed the emergence of various projects and companies concerning transaction tracking on the blockchain, such as Chainalysis (www.chainalysis.com), Etherscan (), and CipherTrace (). Transaction tracking plays an important role for regulators, cryptocurrency exchanges, and investors. For regulators, transaction tracking can assist them to detect fraud behaviors and gangs, follow tainted money, get back stolen cryptocurrencies, and improve the regulatory ecosystem. For example, it is reported that Chainalysis helped the FBI identify two agents on corruption charges of stealing bitcoin from an online drug market operator [15]. Moreover, for cryptocurrency exchanges, transaction tracking improves their capabilities of Know Your Transaction (KYT) and anti-money laundering (AML), and thus helps them better understand the market and protects the rights of law-abiding users. Besides, for ordinary users or cryptocurrency service providers, they can employ the results of transaction tracking to keep their money away from polluted funds and suspicious entities.

Networks are widely adopted as a standard representation of data for understanding interactions between objects in various complex systems [16]. At present, most of the cryptocurrency transaction tracking systems adopt the mechanism combining the network visualization system and manual analysis, such as CoinHolmes (), which is a cryptocurrency AML system of Peckshield (<https://peckshield.com/>). Intuitively, the huge transaction records of cryptocurrencies can be modeled as a network to facilitate analysis. For example, Ethereum accounts in records can be represented as nodes of the network. Nodes are linked if the corresponding accounts have transactions between them. As transaction relationships have direction and multiple transactions between two accounts are expected, it is more comprehensive to model Ethereum transaction records as a directed multigraph, i.e., multi-digraph, rather than a simple graph. Moreover, each edge represents a transaction weighted by the amount and assigned with a timestamp in the Ethereum transaction network.

In the world of cryptocurrency, obtaining the key downstream accounts help track the money flow of criminals. However, money launderers often use a large number of addresses to disperse dirty money during the illegal transfer of assets. As the depth of tracking increases, the constructed network becomes more complex and the downstream addresses grow exponentially. This makes it difficult for regulators and exchanges to recognize dirty money and respond in time to emergencies in time. Fig. 1 shows an example of money laundering by a wealth management wallet TokenStore, which has been exposed as a suspected runaway. Since June 10, 2019, the large-value assets of TokenStore have been divided into new addresses several times layer by layer for intensive money laundering. The money flow starts with account 0x068a,¹ from June 20, 2019. Accounts with different colors denote different layers of the money laundering procedure. For example, account 0xf64d² is one of the 1-hop neighbors of account 0x068a, and account 0x4708³ is one of the 2-hop neighbors. Nodes attached with “ZB”, “Binance”, “OKEx”, and “Huobi” are cryptocurrency exchanges ZB⁴ Binance,⁵ OKEx,⁶ Huobi,⁷ respectively.

From the above example, we can understand that anti-money laundering is a complex financial issue. So far, the academic community has also paid attention to the research issues related to transaction tracking for maintaining financial security and healthy development of blockchain ecosystems. Meiklejohn et al. [17] utilize heuristic clustering to group Bitcoin addresses based on evidence of shared authority, which performs a new perspective for bitcoin de-anonymization and tracking. Yousaf et al. [18] develop techniques to trace cross-ledgers activity by capturing complex transactional behaviors based on data from ShapeShift. Such cross-currency tracing techniques will deter criminals who try to obscure their flow of money. Wu et al. [12] perform the study on the identification of mixing services and tracing illegal transactions in Bitcoin. A systematic analysis with several designed features is conducted to characterize how addresses belonging to mixing services. Wu et al. [19] tackle the transaction tracking task as graph searching the transaction network using Personalized PageRank. They track the fund flow from the source hack addresses with the three kinds of heuristic designs, namely, tracking tendency, weight pollution, and temporal reasoning.

In this work, we try to study the problem of transaction tracking in Ethereum from a network perspective, and investigate the tracking strategies inspired by a random-walk based link prediction model. Specifically, from a microscopic viewpoint, we propose to trace and predict transactions on this huge transaction network via link prediction, which

¹ 0x068ac6ed5efc38a6266261b4486a8907fd7ea15f.

² 0xf46d9659b2e2c5198faf44f5caf5203afb1949b5.

³ 0x474879c0e9e722ae9f14c509f3cf83d49f2d9461.

⁴ <https://www.zb.com/>.

⁵ <https://www.binance.com/>.

⁶ <https://okex.co.kr/kr/view/>.

⁷ <https://www.huobi.com/>.

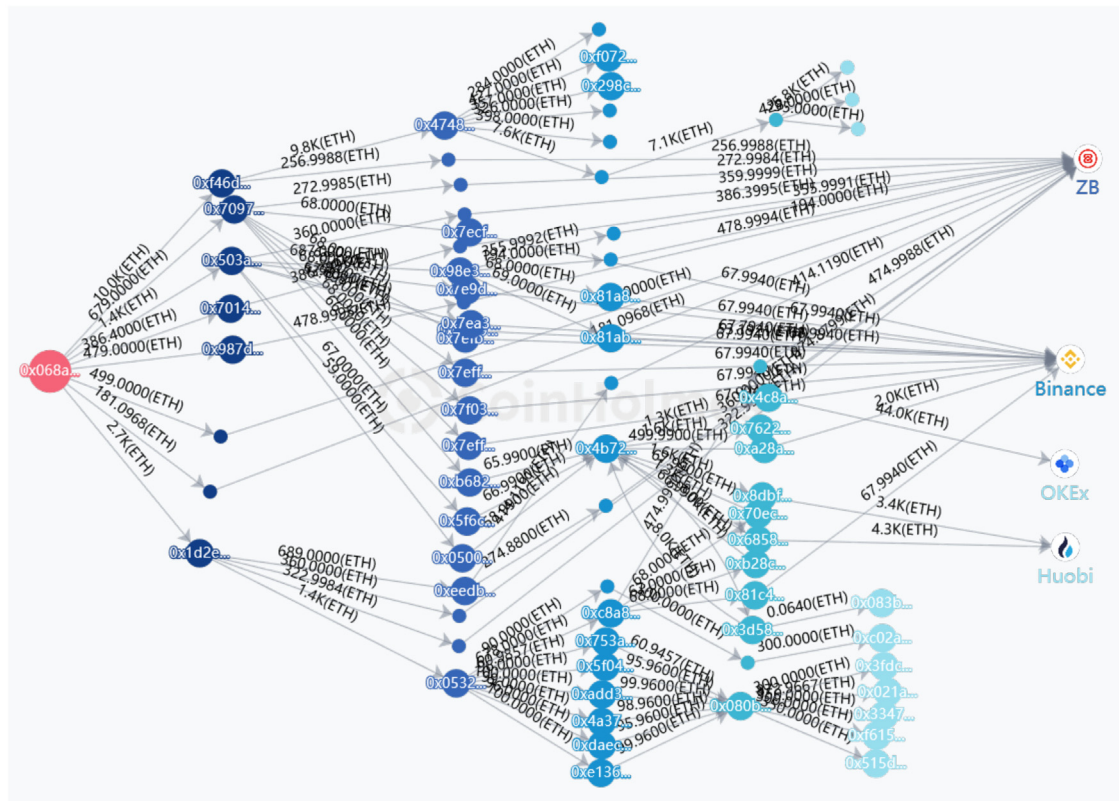


Fig. 1. Money laundering of exposed wallet TokenStore's "running" assets in Ethereum, visualized by CoinHolmes.

is a classic problem in network science. Moreover, from a macroscopic viewpoint, we analyze how the frequency of transactions and the amount being transferred would influence the evolution of the transaction network, leading to better design of transaction tracking methods.

The main contributions of the paper are as follows:

- First, to the best of our knowledge, this is the *first* try to study the transaction tracking problem via link prediction on cryptocurrency transaction networks.
- Second, we propose a random-walk based model and get explainable results about the interplay between transaction features and network evolution of the transaction networks. We explore how two important transaction factors in transaction networks, namely, transaction amount and time, impact the evolution of the Ethereum transaction network.
- Third, we show an application of the transaction tracking technique by conducting phishing account classification, which further demonstrate link prediction can help supervise and monitor the transactions.

Section 2 introduces the related work in the field of blockchain transaction data analysis and network science. Section 3 describes how we collect transaction history from Ethereum, followed by the procedures of data cleaning and preprocessing. In Section 4, we demonstrate the framework for Ethereum transaction tracking, and illustrate how inference can be made on the interplay between transaction information and network evolution via a network-embedding based link prediction model. In Section 5, we further investigate the effectiveness of random-walked based methods. In Section 6, we first collect phishing account transactions, and model the suspicious accounts detection as a binary classification problem, and then use the transaction tracking framework via link prediction to improve the classification performance. We conclude the paper and discuss future directions in Section 7.

2. Related work

In this section, we present two categories of related work. The first one is related to the analysis and mining of blockchain transaction records. The second category includes network evolution theories and link prediction algorithms, from which the network evolution mechanism can be inferred.

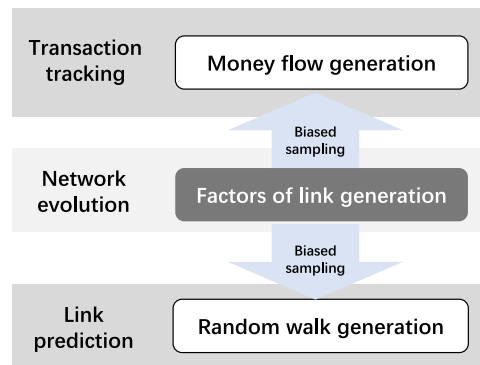


Fig. 2. Relations between transaction tracking, network evolution and link prediction.

2.1. Blockchain transaction analysis and mining

Transaction records are confidential data in financial and business services sectors. The problem of “isolated data island” hinders the research of financial big data mining. Recently, the open nature of blockchain data provides researchers with publicly accessible transaction data, and existing studies on analysis and mining of blockchain transaction records can be categorized into two main groups. One is concerned with the use of appropriate metrics and topological features of complex networks to analyze and characterize blockchain transaction networks [20–25]. The other deals with mining the blockchain transaction via machine-learning-based approaches, and these studies mainly focus on the issues of ownership identification and anomaly detection in blockchain systems [12,14,26–28].

Apart from the ownership identification and detection problem, transaction tracking of cryptocurrencies plays an active role in improving the financial security of blockchain systems. Current research on transaction tracking mainly focuses on tracing the capital flow by inspecting the transaction records, while predicting future transactions is rarely explored. In this paper, we consider the prediction of transactions and attempt to identify the key factors affecting the evolution of Ethereum transaction network via temporal link prediction.

2.2. Network evolution and link prediction

The study of the relationship between various topological structures and properties of complex networks and their dynamic functions have been actively pursued [29]. In particular, network evolution and link prediction are two important topics to which a great deal of research effort has been devoted. In studying the network evolution mechanisms, a typical approach is to investigate whether a network built from some specific growing rules has statistical properties close to those of real networks, such as Erdős–Rényi’s random graph model [30], Watts–Strogatz’s small-world model [31], and Barabási–Albert’s scale-free network model [32].

The purpose of link prediction is to estimate the likelihood of the existence of links between node pairs based on the current network information, e.g., the observed link and node attributes [33]. Existing link prediction methods can be classified into several categories, namely, similarity-based [34,35], probabilistic and maximum likelihood-based [36], and dimensionality reduction-based [37]. In fact, network evolution and link prediction complement each other. On the one hand, indepth analysis of the network evolution mechanism can help researchers design better methods and improve the performance of link prediction. On the other hand, with the theoretical framework and evaluation method of link prediction, evolution models with different factors can be evaluated quantitatively. In addition, link prediction focuses on predicting the generation of new links in the network and network evolution studies the variation trends of the whole network. For instance, Liu et al. [38] discussed a basic framework for predicting the evolution mechanism by link prediction, and revealed the factors affecting the evolution of the Chinese aviation network. The results of link prediction showed that the existence of air routes between two cities is governed by a number of economic factors.

Research questions. We frame our study from the perspective of network science, i.e., the sampling strategies inspired by link prediction. In Fig. 2, we plot the relationships between transaction tracking, network evolution, and link prediction in Ethereum. Since the cryptocurrency transaction is a kind of financial and social behavior [25], the transaction patterns are driven by intrinsic mechanisms, and the transaction networks should exhibit some corresponding evolution rules. Transaction tracking of cryptocurrency consists of money flow generation, which is similar to biased random walk generation in network science. In transaction tracking, the account sequences through which dirty money flows are shaped by particular preferences. For example, hackers are more anxious to move dirty money towards safer places, leading to larger amount transfers. Here, the different preferences of money flows can be modeled as biased random walks. A biased random walk is to sample nodes with preset strategies in networks. The design of the transaction tracking strategies can be quantified as the equation of the selected probability of random walk generations. We seek to answer the following research questions:

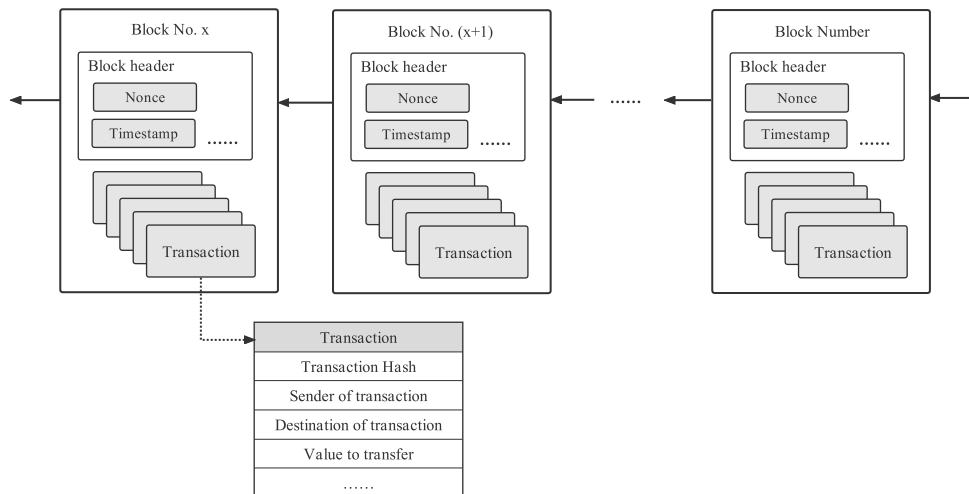


Fig. 3. Ethereum blockchain data structure.

- **RQ1:** Which kind of tracking strategy is more in line with the real cryptocurrency trend, and more suitable for tracking transactions in Ethereum?
- **RQ2:** Can the random-walk based link prediction methods outperform other node-similarity based methods in Ethereum transaction network?
- **RQ3:** Does the proposed link-prediction based tracking framework affect machine learning tasks on Ethereum, such as suspicious account detection?

3. Ethereum transaction records

In this section, we introduce the background of Ethereum transaction data, and describe how we collect and model transaction records from Ethereum.

3.1. Transaction data

As the largest blockchain platform supporting smart contracts, Ethereum provides a Turing scripting language. The Ethereum virtual machine enables Ethereum to handle intelligent peer-to-peer contracts and supports complex and flexible decentralized applications. To facilitate the implementation of smart contracts, Ethereum introduces the concept of “account” and allocates storage space for recording account balances, transactions, codes, etc. Usually, transactions take place between accounts which are represented by addresses composed of prefix “0x”, a common identifier for hexadecimal, concatenated with the rightmost 20 bytes of public key. Here, we use “account” and “address” interchangeably. As shown in Fig. 3, the transaction history is recorded in the blockchain, which consists of data blocks linked in chronological order. So far, the Ethereum blockchain contains more than 8 million blocks.

Etherscan, an Ethereum explorer, provides APIs for users to inquire about the transaction records of given accounts. In this work, we use Etherscan APIs to collect transaction information such as addresses, tokens, prices and other activities taking place in Ethereum, as shown in Fig. 4. From Etherscan, we can collect the addresses of sender and recipient, timestamp, amount value of the transactions.

3.2. Data processing

As mentioned above, since the debut of Ethereum, massive transaction data have been stored in the blockchain. Moreover, the majority of accounts are not always active, and the Ethereum transaction network constructed by the entire transaction records is extremely large and sparse. So much so that it has become increasingly difficult and time-consuming to process the whole Ethereum transaction network for data analysis and mining tasks. To this end, existing work usually focuses on data within certain time intervals or divided by snapshots. For instance, to build and analyze Ethereum transaction graphs, Guo et al. [24] selected 100 thousand blocks with block number from 200,000 to 300,000, and another 200 thousand blocks with block number from 3,000,000 to 3,200,000. Such sampled data is referred to as *time-intensive* transaction data. Although these transactions frequently occur, the corresponding transaction network is usually extremely sparse or even not fully connected, making network analysis and spatial information extraction very difficult.

Overview	Internal Txns	Logs (3)	State
Transaction Hash:	0x30c20cd4f79d54a22b8cb767d39adf7e896afcd5b124d2ea6945d7d3237871bf		
Status:	Success		
Block:	11255356 2 Block Confirmations		
Timestamp:	40 secs ago (Nov-14-2020 10:43:29 AM +UTC) Confirmed within 5 mins:47 secs		
From:	0x567d87b8f2387d51664b49a8c369daae790f4c43		
To:	Contract 0x5acc84a3e955bdd76467d3348077d003f00ffb97 (Forsage.io) TRANSFER 0.05 Ether From Forsage.io To 0x76c5f8a3352d5ea1311fd5d...		
Value:	0.05 Ether (\$22.94)		
Transaction Fee:	0.0026153215 Ether (\$1.20)		
Gas Price:	0.0000000145 Ether (14.5 Gwei)		

Fig. 4. Screenshot of transaction information search on “Etherscan”.

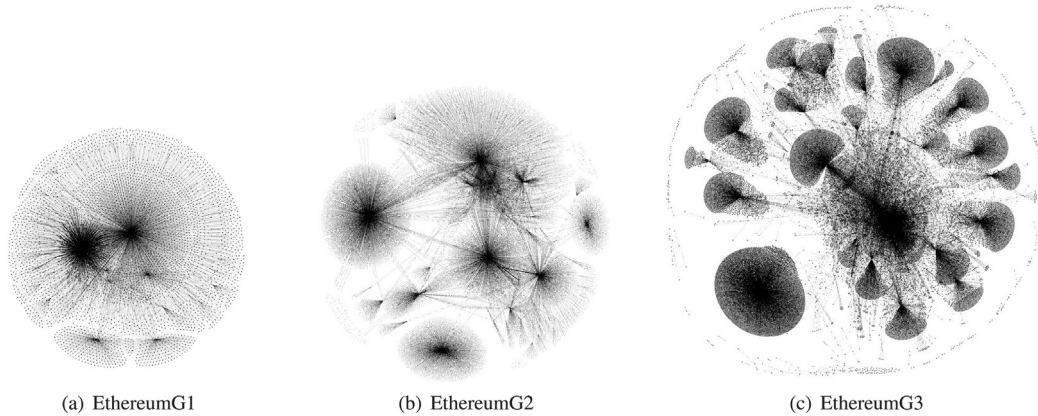


Fig. 5. Visualizations of the network topologies collected from Ethereum transaction datasets.

In this work, we adopt the K -order sampling method proposed in our prior work [39] to obtain *space-intensive* transaction data, whose corresponding transaction network is always connected and can better reserve the local information of the target nodes. Specifically, centered by selected accounts, we sample subgraphs with two parameters, namely, K_{in} and K_{out} , to control the depth of sampling inward and outward from the center, respectively. When $K_{in} = K_{out} = K$, we can view K as the radius of sampled subgraphs [39]. With Etherscan APIs, we recursively query the transaction records until the constructed subgraphs reach the given radius K . During the recursive collection procedure, duplicate transactions are deleted. Using this sampling method, we collect three subgraphs with different size from Ethereum for experiments, and label them as EthereumG1, EthereumG2, and EthereumG3. The datasets presented in this study can be found in online repositories (https://github.com/lindan113/xblock-network_analysis), and is also accessible on our dataset website XBlock (<http://xblock.pro>). The detailed information of datasets are given in our prior work [39]. The visualizations of these graphs are shown in Fig. 5.

3.3. Network construction

The time when a transaction takes place and the amount being transferred are two important attributes of a financial transaction. In most previous blockchain analyses, the transaction records were constructed as a simple network. In this work, in order to incorporate the temporal and weighted information of transactions into consideration, we model the Ethereum transaction records as a *temporal weighted multi-digraph* (TWMDG) which is proposed in [39]. The collected Ethereum data are represented as TWMDG $G(V, E)$, where V denotes the set of accounts and E denotes the set of transactions between accounts. Note that “Multi” indicates multi-transactions between account pairs are modeled as

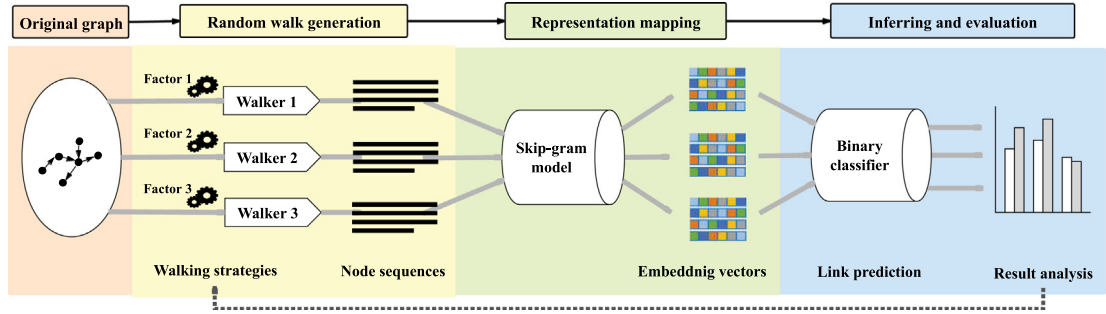


Fig. 6. Framework for inferring the interplay between transaction information and network evolution.

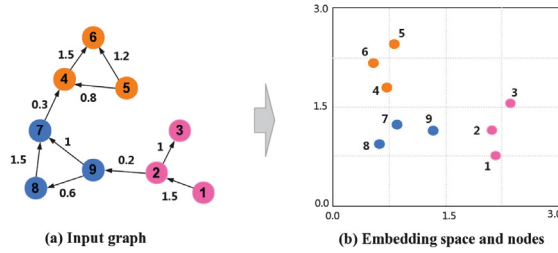


Fig. 7. A toy example of embedding the network into a 2D space.

different edges. Each edge (u, v, w, t) between sender account u and recipient account v is identified by the transaction timestamp t and value w (see Figs. 6 and 9).

4. Method for transaction tracking (RQ1)

4.1. Framework

The transaction records are first converted to a temporal weighted multi-digraph. Then, transaction paths are identified by running random walks to obtain co-occurrence statistics. The original graph is mapped into the embedded space by *representation learning*, which resorts to a dimension reduction method. Finally, by using link prediction methods, we predict the transactions, and infer the impact of transaction features on the evolution of transaction networks by comparing the results derived from different walk strategies.

4.2. Embedding based link prediction

Link prediction is a quantitative tool to investigate the evolution mechanism of networks comprehensively. The money flow in a transaction network is like a sequence of accounts who have transferred *Ether* in chronological order, and therefore the high-order neighborhood should be considered in transaction tracking. However, most of the popular similarity-based link prediction methods can only learn from the one-hop neighborhood. This limitation may lead to low accuracy of link prediction task. High-order proximity explores k -step ($k \geq 3$) relations and captures the similarity between nodes close to each other in the network [40].

Network embedding, also known as network representation learning, is a dimension reduction technique for obtaining an effective representation of large networks that preserves the node neighborhood structure [41]. As shown in Fig. 7, similar nodes in the original network should have similar node vectors in the embedded low-dimensional space.

Recently, random-walk based network embedding has been widely used to deal with link prediction tasks in various practical networks, e.g., DeepWalk [42], node2vec [37], and *continuous time dynamic network embedding* (CTDNE) [43]. Besides, high-order proximity can better be preserved in the biased random walk. Temporal-random-walk based network embedding capture the information of high-order neighborhood by the walker continuously sampling nodes in chronological order. In this paper, for the Ethereum transaction network modeled as temporal weighted multi-digraph, we propose a random-walk based node embedding framework by considering different walking strategies based on some key transaction features. In this way, the original network is transformed into different collections of node sequences by sampling truncated random walks. The occurrence frequency of a node on short random walks and its content reflect the proximity between them. Therefore, the obtained node sequences with different walk preferences would result in various node representations via an optimization procedure. The proposed embedding method consists of the following two parts:

Algorithm 1 Temporal Walk**Input:** TWMDG $G = (V, E)$, start node u , walk length l **Output:** $walk_n$

```

1: Let  $v_1 = u$ , initialize  $walk_n$  to  $[v_1]$ ,  $walk_e$  to  $\emptyset$ 
2: Randomly sample first edge  $e_1$ , append  $e_1$  to  $walk_e$ 
3: Let  $v_2 = dst(e_1)$ , append  $v_2$  to  $walk_n$ 
4: for  $i = 2$  to  $l - 1$  do
5:    $cur_n = walk_n[-1]$ ,  $cur_e = walk_e[-1]$ 
6:    $e_i = GetNextEdgeWithStrategies(cur_n, cur_e)$ 
7:   Append  $e_i$  to  $walk_e$ ,
8:   Let  $v_{i+1} = dst(next_e)$ , append  $v_{i+1}$  to  $walk_n$ ,
9: end for
10: return  $walk_n$ 

```

Algorithm 2 Temporal Weighted Multidigraph Embedding**Input:** Temporal Weighted Multidigraph $G = (V, E)$, dimensions d , walks per node r , walk length l , window size k **Output:** $\Phi(v)$ for $\forall v \in V$

```

1: Initialize set of temporal walks  $TW$  to  $\emptyset$ 
2: for  $iter = 1$  to  $r$  do
3:   for all nodes  $u \in V$  do
4:      $walk_n = TemporalWalk(G, u, l)$ 
5:     Append  $walk_n$  to  $TW$ 
6:   end for
7: end for
8:  $\Phi = StochasticGradientDescent(k, d, TW)$ 
9: return  $\Phi \in \mathbb{R}^{|V| \times d}$ 

```

1. A **random walk generator** provides a set of walks with temporal constraints and in accordance with given flexible biased strategies. We adopt the random walk generator from [39], and the pseudocode for temporal walk generation is given in Algorithm 1. In the function *GetNextEdgeWithStrategies*, we select the next edge for the temporal walk. In Tables 1 and 2, we describe different sampling biases by formulating the selection probability for each temporal successive edge $e \in L_t(v_i)$ [39], considering transaction time and amount, respectively. The previous walk is represented as e_{i-1} with timestamp $t = T(e_{i-1})$, and the current node is represented as v_i , where function T gets the timestamp of transaction e_{i-1} . The function W gets the value transferred of transaction e_{i-1} . We denote $\eta_- : \mathbb{R} \rightarrow \mathbb{Z}^+$ as a function that maps the timestamps of edges to positive integers of descending order, and let $\eta_+ : \mathbb{R} \rightarrow \mathbb{Z}^+$ be a function that maps the timestamps (or amount) of edges to positive integers of ascending order. For example, if the current temporal successive edges e_a , e_b and e_c have transaction values 1ETH, 10ETH and 100ETH, respectively, then we have $\eta_+(e_a) = 1$, $\eta_+(e_b) = 2$, and $\eta_+(e_c) = 3$. If e_a , e_b and e_c have transaction timestamp 1526666690, 1499026232 and 1497221976, respectively, then we have $\eta_-(e_a) = 1$, $\eta_-(e_b) = 2$, and $\eta_-(e_c) = 3$. A detailed explanation of each sampling strategy will be given in Section 4.4.
2. An **optimization procedure** is applied to achieve node embeddings and adopts a “skip-gram” learning algorithm for updating the network [44,45]. The procedure essentially takes a maximum likelihood optimization approach. Given the sampling results of temporal random walks, we formulate the task of learning time and weight dependent graph embedding in a TWMDG as an optimization problem. Using the skip-gram model, node representations are learned by maximizing the probability of observing the neighborhood of a node conditioned on its embedding.

The pseudocode for the above temporal weighted multidigraph embedding is given in Algorithm 2. The output is an embedding function $\Phi : V \rightarrow \mathbb{R}^d$ (with node embedding dimension $d \ll |V|$) which preserves the original network information including node proximity. This embedding method is scalable to large networks. The time complexity of random walk generator is $\mathcal{O}(r|V|l)$ and the time complexity of optimization procedure is $\mathcal{O}(|V| \log |V|)$, where r denotes walks per node, $|V|$ denotes the number of nodes and l denotes walk length.

Then, we obtain the embedded features of a directed link from nodes v_i to v_j by concatenating their node embeddings, i.e., $F_{i,j} = [\Phi(v_i), \Phi(v_j)]$. If $i \neq j$, $F_{i,j} \neq F_{j,i}$. Link prediction can be viewed as a supervised binary classification problem, i.e., whether there is a link between a given pair of nodes or not. Thus we train *support vector machine* (SVM), which is a classic binary classifier, to classify the links are exist in the test set. In summary, different walking strategies generate different node sequences with different co-occurrence statistics, and result in various node representations with different proximity.

Table 1

Sampling biases for different walking strategies with timestamp (time elapsed in transaction) of edges considered. $T_{\max}$ is the largest timestamp of current network.

Sampling criterion	Walking strategies	Defining equation
From WS_4 to WS_1 , the random walks tend to sample the successive transactions with a shorter time interval.	WS_1	$P_T(e) = \frac{T_{\max} - T(e)}{\sum_{e' \in L_T(v_i)} (T_{\max} - T(e'))}$
	WS_2	$P_T(e) = \frac{\eta_-(T(e))}{\sum_{e' \in L_T(v_i)} \eta_-(T(e'))}$
	WS_3	$P_T(e) = \frac{1}{ L_T(v_i) }$
	WS_4	$P_T(e) = \frac{\eta_+(T(e))}{\sum_{e' \in L_T(v_i)} \eta_+(T(e'))}$

Table 2

Sampling biases for different walking strategies with edge weight (transaction amount) considered.

Sampling criterion	Walking strategies	Defining equation
From WS_7 to WS_5 , the random walks tend to sample the successive transactions with a larger transaction amount.	WS_5	$P_W(e) = \frac{W(e)}{\sum_{e' \in L_T(v_i)} W(e')}$
	WS_6	$P_W(e) = \frac{\eta_+(W(e))}{\sum_{e' \in L_T(v_i)} \eta_+(W(e'))}$
	WS_7	$P_W(e) = \frac{1}{ L_T(v_i) }$

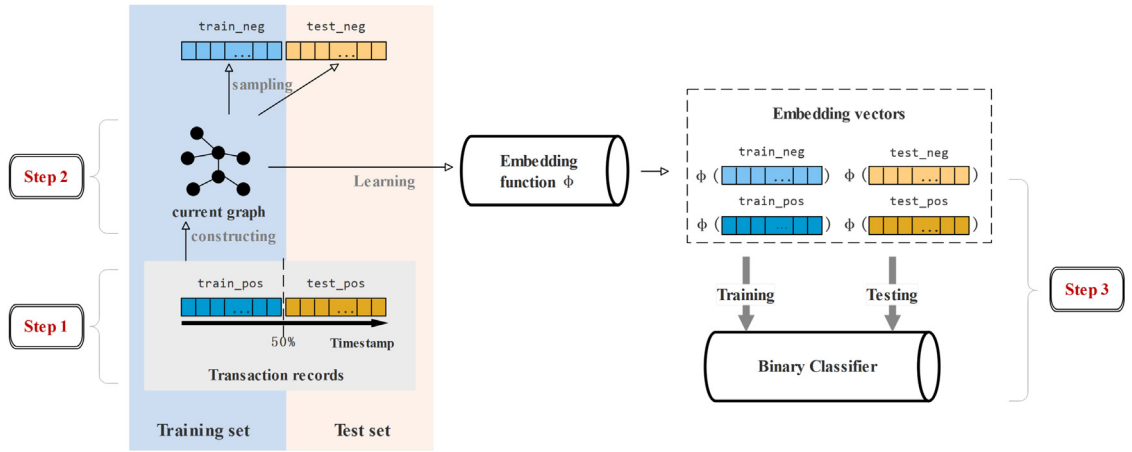


Fig. 8. The main steps of temporal link prediction. $train_pos$, $test_pos$ are the positive links of the training set and test set, and $train_neg$, $test_neg$ are the negative links of the training set and test set, respectively.

Different network representations may show different link prediction performance under a binary classifier. By comparing link prediction results, we can identify the walking strategy that achieves a better sampling and prediction of Ethereum transactions, and further infer the key factors influencing the evolution of the whole system.

4.3. Evaluation of temporal link prediction

In conventional link prediction experiments where links have no timestamp, the observed links of the network are usually randomly divided into training and test sets. For the problem of link prediction in a temporal network considered here, it is more reasonable to split the train-test set according to the timestamp and use the existing links to predict the occurrences of future links.

The main steps of the temporal link prediction, as illustrated in Fig. 8, are:

1. Sort all the collected edges from transaction records according to their timestamps and assume 50% of the earlier edges as the known edges, which constitute the current network.
2. Perform learning of node representations of the current network $\phi(v)$ for $\forall v \in V$ via graph embedding methods.
3. For the binary classifier, the 50% links with earlier timestamps in the original network act as positive samples of the training set, and the remainder (the 50% links with later timestamps) are treated as the positive samples of the test set. Besides, we randomly sample an equal number of node pairs with no link as negative samples of the training set and the test set.

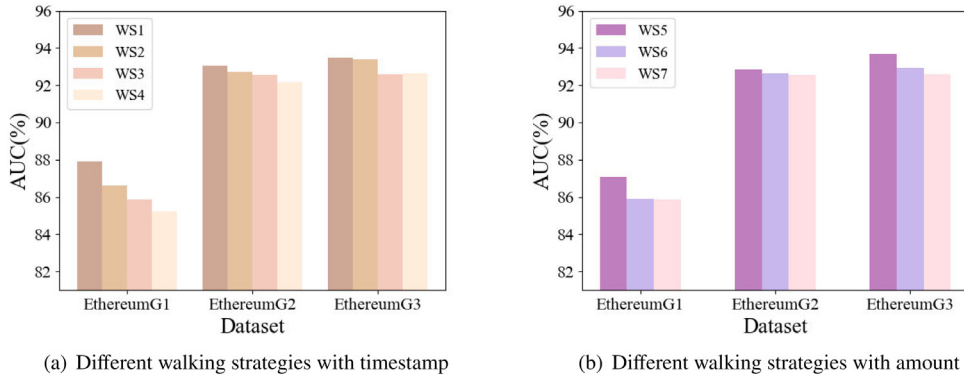


Fig. 9. Link prediction results in terms of AUC considering different walking strategies. (a) considering the transaction time, (b) considering the transaction amount.

4.4. Results and discussions

4.4.1. Transaction information

First, we analyze the effect of transaction information, specifically the *time elapsed between successive transactions* and the *amount being transferred*, on the evolution of Ethereum transaction network. Consider a random walk which is currently stopping at node v_i . The next node of the random walk is decided by selecting a temporally valid edge of node v_i . As mentioned before, this edge is selected from the set of temporal successive edges, $L_t(v_i)$, using a particular sampling bias. Here, we employ several biased random walk sampling methods based on the *temporal* and *amount* information for link prediction experiments, aiming to demonstrate how these two factors influence the generation of new transactions.

Effects of transaction frequency. First, we investigate the effects of temporal information by considering four kinds of walking strategies according to transaction time information, as explained below:

- Walking strategy WS₁: In this strategy, the selection probability of the next transaction is linearly dependent on the time interval between the next transaction's timestamp $T(e)$ and the largest timestamp $T_{\max}$ of the current network.
- Walking strategy WS₂: As in WS₁, an earlier transaction has a higher probability to be selected. However, WS₂ strategy employs a descending rank function η_- which maps the timestamps of edges to positive integers of descending order. The selection probability of the next step is linearly dependent on the (descending) rank of the timestamp $\eta_-(T(e))$.
- Walking strategy WS₃: In this strategy, each candidate transaction has an identical probability to be selected.
- Walking strategy WS₄: This strategy is the opposite of WS₂, and employs an ascending rank function η_+ which maps the timestamps of edges to positive integers of ascending order. The selection probability of the next transaction is linearly dependent on the (ascending) rank of the timestamp $\eta_+(T(e))$.

The essence of the various strategies is to adjust the effects of the time interval between successive edges on sampling random walk paths, which may influence the proximity between nodes in the graph. We can see from Table 1 that, from WS₁ to WS₄, the random walks tend to sample the successive transactions with longer time intervals. Fig. 9(a) compares the performance of link prediction under different walking strategies in terms of AUC. The link prediction algorithm based on WS₁ outperforms the rest.

On transaction tracking of cryptocurrencies, we can infer that in an Ethereum transaction network, the generation of new transactions is related to the time elapsed, or frequency of transaction in the reciprocal sense. Thus, new transactions are more likely to take place between node pairs with more frequent transactions in the path. Therefore, when tracking the money flow among Ethereum accounts, we should pay more attention to those transaction paths having a shorter time interval.

Effects of transaction amount. Next, we investigate the effects of edge weight information (i.e., transaction amount) on the link prediction experiments. Considering the transaction amount, each candidate edge $e \in L_t(v_i)$ is assigned a selection probability $P_W(e)$ under different walking strategies whose calculation methods are given in Table 2. Detailed explanations are as follows.

- Walking strategy WS₅: In this strategy, the probability of selecting a candidate edge is proportional to its edge weight, i.e., the transaction amount $W(e)$.
- Walking strategy WS₆: In this strategy, an edge with higher weight has a higher probability to be chosen, and a linear mapping function (i.e., ranking function) is adopted to weaken the impact of edge weight information compared with WS₅. Specifically, the selection probability of the next transaction is linearly dependent on $\eta_+(W(e))$.
- Walking strategy WS₇: Each candidate transaction has an identical probability to be selected.

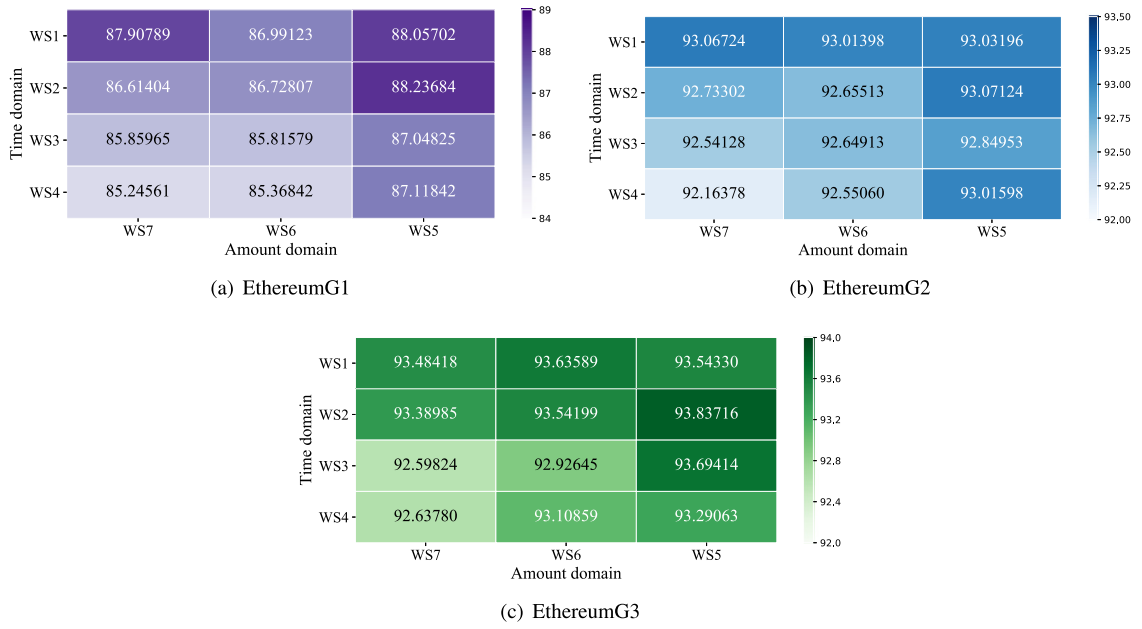


Fig. 10. Link prediction results in terms of AUC considering both the transaction time lapse and amount.

From WS₅ to WS₇, the impact of edge weight (transaction amount) gradually weakens. The results given in Fig. 9(b) demonstrate that the link prediction model based on WS₅ attains the best performance, implying that the likelihood of occurrence of a future transaction is positively related to the transaction amount between two accounts. In most cases, the transactions with larger values infer that the two corresponding accounts have a closer relationship. Thus, we should pay more attention to those transaction paths having a larger transaction amount.

Effect of time and amount. We combine both the temporal and weight factors, and investigate how they influence transaction evolution. The combined influence can be represented by the following probability of selecting edge e :

$$P(e) = \frac{P_T(e)^\alpha P_W(e)^{(1-\alpha)}}{\sum_{e' \in L_t(v_i)} P_T(e')^\alpha P_W(e')^{(1-\alpha)}} \quad (1)$$

where α is a tuning parameter varying from 0 to 1. Here $\alpha = 0.5$ is the default value, which corresponds to a balanced temporal and weight influence. Note that $P_T(e)$ denotes the selecting probability for WS₁ to WS₄, and $P_W(e)$ denotes the selecting probability for WS₅ to WS₇.

The results of all possible combinations of walking strategies are shown in Fig. 10. We observe that the link prediction model tends to achieve better performance from bottom to top, and from left to right. Moreover, when the transaction amount is taken into account, especially with WS₅ (the rightmost column), the difference between combining different time-lapse strategies is less obvious. The results further verify that both the “temporal” and “amount” factors impact the Ethereum transaction interaction, and imply that the amount being transferred has a greater influence than the time elapsed between transactions on the network evolution.

5. Transaction tracking via similarity-based link prediction (RQ2)

As explained in Section 4.2, random walk based embedding methods can effectively embed a large sparse network into a low dimension space while preserving the node similarity property. However, comparing with traditional link prediction methods, the effectiveness of our embedding based framework on tracking transactions and network evolution is yet to be verified. Here, to study the competitiveness of our embedding based framework, we conduct experiments of various similarity-based link prediction methods.

As presented in Section 2.2, a specific link prediction model corresponds to a network evolution mechanism. Usually, a traditional link prediction model is accompanied with an explicit node-similarity measurement for node pairs, which indicates how the new connections are generated between unconnected node pairs in the network. Given a link prediction method, we first calculate the similarity value for each unconnected node pair. It is assumed that node pairs with higher similarity values are more likely to generate new links in the future. The similarity indices can be grouped into the node-dependent indices and path-dependent indices:

- Node-dependent similarity indices

- Common Neighbors (CN) [34] of nodes u and v is defined as

$$|\Gamma(u) \cap \Gamma(v)|,$$

where $\Gamma(u)$ and $\Gamma(v)$ denote the set of neighbors of u and v , respectively. The link prediction method based on CN assumes that nodes u and v are more likely to be connected if they have more common neighbors.

- Jaccard Coefficient [46] of nodes u and v is

$$\frac{|\Gamma(u) \cap \Gamma(v)|}{|\Gamma(u) \cup \Gamma(v)|},$$

which considers the ratio of common neighbors among all the neighbors rather than the absolute value of common neighbors.

- Resource Allocation Index (RA) [47] of nodes u and v is defined as

$$\sum_{w \in \Gamma(u) \cap \Gamma(v)} \frac{1}{|\Gamma(w)|},$$

which is proposed from the resource allocation perspective. Each node is assumed to have a unit of resource being distributed equally to its neighbors.

- Adamic-Adar Index (AA) [48] of nodes u and v is

$$\sum_{w \in \Gamma(u) \cap \Gamma(v)} \frac{1}{\log |\Gamma(w)|}.$$

The idea of AA is that the common neighbors with lower degree contribute more than those with higher degree for link prediction.

- Path-dependent similarity indices

- Local Path Index (LP) [49] takes the third-order neighborhood into consideration, except for Common Neighbors (second-order neighborhood). The LP index of nodes u and v is defined as

$$(A^2 + \epsilon A^3)_{uv},$$

where A is the adjacency matrix, ϵ is the parameter used to control the effect of the three order path and here we set $\epsilon = 0.1$.

- Katz Index [35] of nodes u and v aggregates all the paths between them, assigning shorter paths with smaller weights and longer paths with larger weights. The Katz Index of nodes u and v is defined as the

$$(\beta A + \beta^2 A^2 + \beta^3 A^3 + \dots)_{uv}$$

where β is a damping factor which is set as 0.01 in our experiments.

- Random Walk with Restart Index (RWR) [50] of nodes u and v is defined as

$$q_{uv} + q_{vu},$$

where q_{uv} is the probability that a random walker starting from node u eventually reaches v . Let $(1 - c)$ be the probability of the random walker returning to the previous node, and let P be the transition matrix. Then, we have $\vec{q}_u(t + 1) = cP^T \vec{q}_u(t) + (1 - c)\vec{e}_u$. In our experiments, we set $c = 0.1$.

- Average Commute Time Index (ACT) [51,52] assumes that two nodes u and v are more similar if the value of ACT between them is smaller, and ACT of nodes u and v is defined as

$$\frac{1}{L_{uu}^+ + L_{vv}^+ - 2L_{uv}^+},$$

where L^+ is the pseudo-inverse of the Laplace matrix of the network.

Based on the similarity indices, we take the unconnected node pairs with higher similarity values as positive samples, and evaluate the performance by AUC. The AUC results of different similarity indices are shown in Table 3. We see that the performances of node-dependent similarity indices, i.e., CN, Jaccard, RA and AA, are very similar on the three datasets. Among the path-dependent similarity indices, LP shows competitive performance in dataset EthereumG1, but performs not so well in the other two datasets. RWR and ACT are memory-consuming for dataset EthereumG3, which indicates that RWR and ACT are not suitable for large-scale transaction networks. As shown in Table 3, the proposed network embedding (NE) based methods outperform traditional link prediction methods, which confirms that network embedding methods with flexible walking strategies are able to learn similarity between nodes more effectively. We thus conclude that the embedding based framework is effective for transaction tracking of large-scale Ethereum networks.

Table 3

Link prediction results in terms of AUC for similarity-based and proposed network embedding based methods. “NE (uniform)” denotes the proposed network embedding based methods with walking strategies WS_3+WS_7 , and “NE (best)” gives best results among combinations of walking strategies.

Dataset	Area under curve (AUC) (%)									
	CN [34]	Jaccard [46]	RA [47]	AA [48]	LP [49]	Katz [35]	RWR [50]	ACT [51]	NE (uniform)	NE (best)
EthereumG1	73.33	73.33	73.33	73.33	86.32	72.81	79.12	48.42	85.86	88.24
EthereumG2	87.67	87.67	87.67	87.67	87.30	87.67	89.21	58.27	92.54	93.07
EthereumG3	92.29	92.29	92.29	92.29	91.95	92.04	–	–	92.60	93.84

Table 4

Part of the labeled accounts reported by Etherscan. “#” denotes “number of”, and “TX” denotes “Transactions”.

Accounts	Name tag	#TX
0x21f74c6bbc1e3ab9f0205e12de3a9daa14351aed	Fake_Phishing702	1,619
0x3cbd2e6143f057bd49ffb4c7058217a5900c35d3	Fake_Phishing311	1,587
0x6aef2850c2ce2ef583ae8a36be3357080608bdc3	Fake_Phishing171	924
0x1451dbe4304a60b5b25b4d98b355effd440417af	Fake_Phishing186	694
0x474057adf42f9f955e86aa1142740f9d7763e41e	Fake_Phishing710	681
0xb00548770ea87482f024cd4292a975fea72e8f01	Fake_Phishing235	644
0x4bf722014e54aeab05fcf1519e6e4c0c3f742e43	Fake_Phishing250	621
0x9ef89d10d8c77755bfe263903966eebb73f721e8	Fake_Phishing252	608
0x39b6ce3428068ce97db218fd2f8df6909b331308	Fake_Phishing359	595
0x4f00b95c625c6d6ef72748d78c750fd6c84a8b85	Fake_Phishing337	581

6. Application of transaction tracking (RQ3)

As we have mentioned in Section 1, criminal activities on blockchain have become more frequent and rampant in recent years. According to the Chainalysis 2020 Crypto Crime Report,⁸ 2019 was the biggest year for cryptocurrency scams—Scammers brought in 4.3 billion USD worth of cryptocurrency from millions of victims. Ether has long been known as the cryptocurrency of choice for scams, and the phishing scam is one of the most common types of Ethereum scams. Usually, the target victims receive phishers’ email or other communication that tricks them into sharing personal financial information that allows access to their Ethereum wallets. According to the report,⁹ illegal revenue from phishing scammers exceed 3 million USD in just one month in the early 2018. These shows that detecting and tracking phishing scams is an urgent problem in the Ethereum ecosystem.

Cryptocurrency networks are continuously growing over time. In the evolving and dynamic Ethereum transaction network, the network information we get may be incomplete, which may result in the low accuracy of illicit account detection task in the cryptocurrency world. In Section 4, we have demonstrated the framework for Ethereum transaction tracking based on a link prediction model, and the good link prediction model can also be used to describe the evolution mechanism of complex networks. We assume that with the supplement data predicted by evolution mechanism model, missing information can be added to detect illicit activities in the work of cryptocurrency. Therefore, in this section, we first model the suspicious accounts detection as a binary classification problem on the transaction network, and leverage F1-score as the evaluation criterion. Then we use the link-prediction based framework to improve the classification performance, by supplementing the unknown or missing information of the Ethereum transaction network. We conduct experiments on the real Ethereum transaction data to evaluate whether the application of link prediction model can enhance the performance of suspicious account detection.

Step 1: Obtain the local transaction network of target accounts. We use K -order sampling method [39] to obtain the transaction data of 445 phisher accounts labeled in the “Label Word Cloud” service provided by Etherscan.¹⁰ Table 4 shows 10 of the accounts tagged as phishing scams by Etherscan. Besides, we randomly select unlabeled accounts as the negative samples to train our classification model. The transaction data of 890 target accounts are modeled into a temporal weighted multidigraph with 86,622 nodes and 262,740 edges.

The sampling depth of inward and outward from the center account are set to $K_{in} = 1$, $K_{out} = 3$. The reason of the different K_{in} and K_{out} is the cryptocurrency flow characteristic of suspicious accounts. Here we show an example of a typical transaction network centered on a phishing account 0xc440.¹¹ The direct incoming transactions of 0xc440 is the process of obtaining funds from victims. The outgoing transactions from a phishing account can be seen as the money laundering process, based on the assumption that a scammer has hardly a reason to transfer the funds to an account that has no relationship with, thus the outgoing transaction neighborhood may exist a potential scamming gang. One of the

⁸ <https://go.chainalysis.com/2020-Crypto-Crime-Report.html>.

⁹ <https://go.chainalysis.com/2019-Crypto-Crime-Report.html>.

¹⁰ etherscan.io/accounts/label/phish-hack.

¹¹ 0xc440015b2311b109949309a81624fb4dbadecdd0.

Table 5

The F1-score (%) of phishing account classification on the collected Ethereum transaction records.

Test ratio	50%	40%	30%	20%	10%
F1-score	79.55	80.22	80.60	80.79	80.67

Table 6The F1-score (%) of phishing account classification before/after random attack. The ratio of removals $\theta\%$ indicates how many edges removed from the current network.

Removals ($\theta\%$)	Test ratio				
	50%	40%	30%	20%	10%
10%	78.70	78.88	79.7	79.89	81.12
20%	77.17	77.02	77.9	76.07	79.10
30%	74.88	75.67	75.28	75.28	75.28
40%	72.36	72.13	71.91	71.24	73.71

outgoing neighbors of the phishing account 0xc440 is 0x008c,¹² which is unlabeled in Etherscan, but 0x008c send Ether to account 0x4985,¹³ which is labeled as “Fake_Phishing982” in Etherscan. Another example is that the phishing account 0xc440 transferred Ether to a bridge account 0x5902,¹⁴ which is also unlabeled, but the account 0x5902 further transfer Ether to a Binance wallet 0x3f5c¹⁵ for cryptocurrency exchange. Above all, tracing the flow of funds, the accounts in the path from the phishing account may be suspected of being accomplices, as long as the endpoint is an exchange wallet. We set 3 as the outward sampling depth according to the threshold setting of a taint analysis in [53].

Step 2: Account feature learning and suspicious accounts detection. We obtain the latent features of accounts by a network embedding method T-EDGE [54], mapping each target account to a 128 dimensional feature vector. We randomly select parts of the accounts as the test set, and remain the rest as the training set, respectively. We classify the phisher/unlabeled accounts by training a SVM model, and report the F1-score in Table 5.

Step 3: Random attacks on the current transaction network. The incomplete topological information of Ethereum network may do harm to phishing account classification. The concern is that the blockchain explorer Etherscan discourages automated crawlers and the provided API claims that it can only download the last 10,000 transactions [55]. To simulate this situation, we randomly remove $\theta\%$ of the edges from the transaction network, and then repeat the experiments in Step 2. The experimental results are shown in Table 6 and we compare them with the result in Table 5. As the ratio of removals increased from 0% to 40%, the F1-score of 50% test ratio is decreasing from 79.55% to 72.36%, and the F1-score of 10% test ratio is decreasing from 80.67% to 73.71%. We can draw a conclusion that the incomplete structural information results in a poorer performance of phishing account classification. In most cases, networks with larger ratio of removals gets lower F1-score.

Step 4: Link prediction on the incomplete transaction network. We conduct the experiment mentioned in Section 4.3 to train a temporal link prediction model. As mentioned in Section 4.3, we use 50% of the transactions with smaller timestamps for temporal link prediction training. The temporal link prediction result on the test set in terms of AUC is 89%, with the node embedding dimension $d = 128$, the size of window $k = 4$, the length of walk $l = 10$, and walks per node $r = 20$ for the random-walk based link prediction. Then we randomly sample L unconnected account pairs and predict their positive and negative based on the trained temporal link prediction model. The predicted positive samples are added to the current transaction network, and we denote pos as the number of predicted positive links. Experiment results on the extended dataset are shown in Table 7 and we compare them with the result in Table 5. It makes sense that as L grows tenfold, pos grows approximately tenfold. As L increased from 0 to 500,000, the F1-score of 50% test ratio is increasing from 79.55% to 84.31%, and the F1-score of 10% test ratio is increasing from 80.67% to 86.52%. Our experiments confirm that transaction tracking via link prediction effectively enhances the ability to detect the suspicious accounts in Ethereum.

7. Conclusion

In this paper, we study the transaction tracking problem in Ethereum from a network science perspective and infer on the interplay between transaction features and network evolution. Specifically, we present a biased random-walk based link prediction framework to study transaction tracking. To investigate the impact of transaction frequency and amount on the evolution of transactions, we design a random walk generator with various walking strategies to examine the temporal network embedding process. Simulation results on three Ethereum transaction networks reveal that taking both the time lapses between transactions and the amount being transferred into consideration, the link prediction can achieve

¹² 0x008c61e96f6ac0042aef8eb14be5720414bf5df6.

¹³ 0x4985a8d068a9cac4b81faaafb8d4516ae46dd768.

¹⁴ 0x59027d7292ae7f48fb80f562a946d8115f98608e.

¹⁵ 0x3f5ce5fbfe3e9af3971dd833d26ba9b5c936f0be.

Table 7

The F1-score (%) of phishing account classification after link prediction. L is the number of links in the test set for link prediction. $|pos|$ is the number of predicted positive links added to the current transaction network.

L	$ pos $	Test ratio				
		50%	40%	30%	20%	10%
500	52	80.63	80.39	81.42	81.57	83.37
5,000	429	80.90	81.07	81.72	81.12	81.57
50,000	4,266	81.53	83.09	82.62	82.92	85.17
500,000	41,970	84.31	83.37	84.34	83.37	86.52

better performance. Moreover, by comparing the results of different sampling biases, we can infer that in the Ethereum transaction network, occurrences of new transactions are related to the frequency and amount of previous transactions. In particular, new transactions are more likely to occur between account pairs who have common neighbors with higher transaction frequency and amount. Our embedding based framework is also found to be more effective than traditional methods of link prediction and transaction tracking. In addition, we use the link-prediction based transaction tracking technique to improve the phishing classification performance by supplementing the unknown or missing information of Ethereum transaction network, which further verify how transaction tracking can help trace transactions and identify frauds.

It is thus valuable to further explore the evolution mechanisms of different cryptocurrencies, and identify the similarities and differences in these mechanisms, which are relevant to cross-platform transaction tracking. Our future work will include investigation of the influence of other account features on the evolution mechanism by refining the link prediction model.

CRediT authorship contribution statement

Dan Lin: Conceptualization, Methodology, Investigation, Data curation, Writing – original draft. **Jiajing Wu:** Writing – review & editing, Supervision, Funding acquisition. **Qi Xuan:** Writing – review & editing. **Chi K. Tse:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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